WEEK 8-SVM

ML

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Section:F

1.Objective

To implement Support Vector Machine (SVM) classifiers using Linear, Radial Basis Function (RBF), and
Polynomial kernels on the Moons and Banknote datasets, evaluate their performance using classification reports, visualize decision boundaries, and compare soft-margin and hard-margin models

2. Methodology

1. Loaded the required Python libraries and datasets.
2. Split the datasets into training and testing sets.
3. Scaled the features using StandardScaler.
4. Trained SVM classifiers with Linear, RBF, and Polynomial kernels.

5. Predicted test labels and examined the classification reports.
6. Plotted decision boundaries for the models.
7. Compared a soft-margin model (C = 0.1) with a high-C model (C = 100)

3. Analysis Questions — Moons Dataset

Q1. What can you infer about the Linear kernel's performance?

The Moons dataset has a curved pattern. A Linear kernel creates a straight decision boundary, so it may not separate the two classes well. The classification report and plot should be used to confirm its performance.

Q2. Compare the RBF and Polynomial decision boundaries.

The RBF and Polynomial kernels can create curved decision boundaries. The RBF kernel can fit flexible non-linear patterns, while the Polynomial kernel creates a boundary based on a polynomial function. Their plots and test scores show which one works better for this dataset.

4. Analysis Questions — Banknote Dataset

Q1. Which kernel appears to be most effective?

The kernel with the highest test accuracy and F1-score in the classification reports is the most effective.

Compare Linear, RBF, and Polynomial results to identify the best one for this run.

Q2. Why might the Polynomial kernel perform worse here than on the Moons dataset?

The Polynomial kernel may not match the pattern of the Banknote dataset as well. Different datasets have different patterns, so a kernel that works well on the Moons dataset may not work
equally well on the Banknote
dataset.

5. Analysis Questions — Soft Margin and Hard Margin

Q1. Which model produces a wider margin? The Soft Margin model (C = 0.1) generally produces a wider margin than the model with C = 100.

Q2. Why does the Soft Margin model allow mistakes? What is its main goal? The Soft Margin model allows some mistakes because

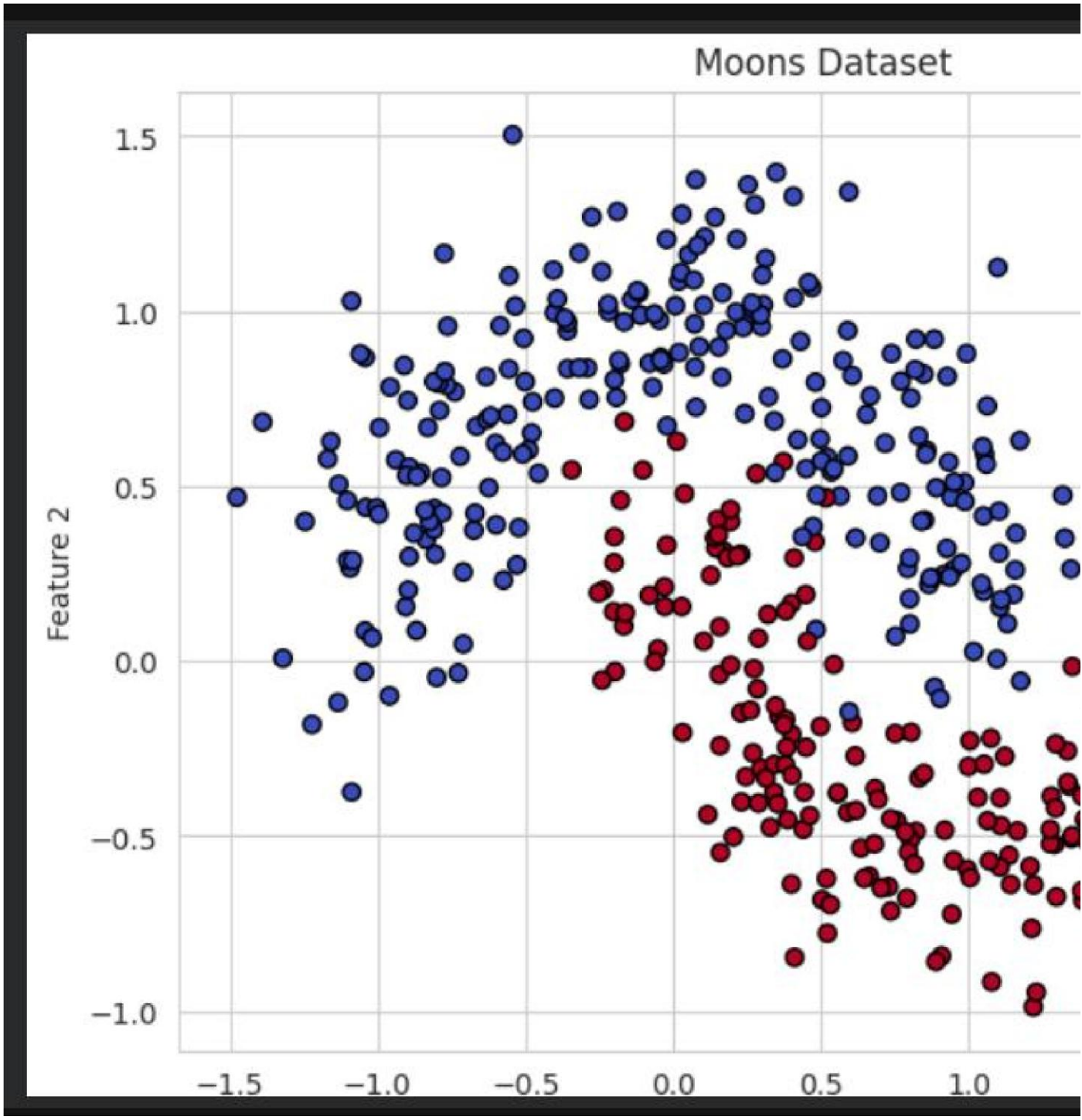
it uses a low C value. Its goal is to maintain a wider margin and potentially perform better on new, unseen data.

Q3. Which model is more likely to overfit?
The model with C = 100 may be more likely to overfit because it penalizes training errors strongly and tries harder to classify training points correctly. This is not guaranteed, so test performance should also be checked.

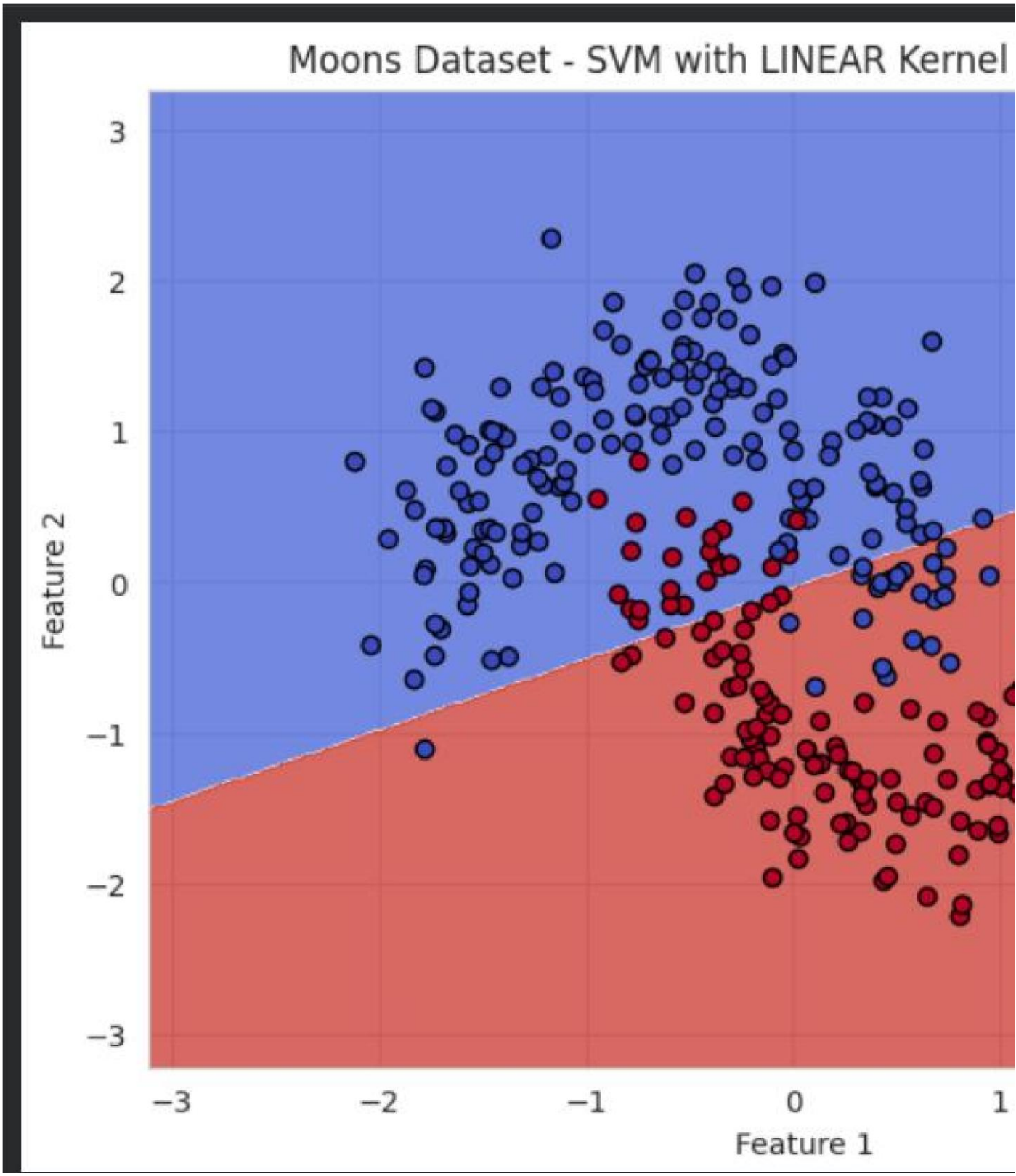
Q4. Which model would you trust for new data, and which C value would you start with?
I would generally start by testing a low C value such as 0.1 because real-world data can contain noise. A low
C allows some training errors and may generalize better. The final choice should be based on validation or test results

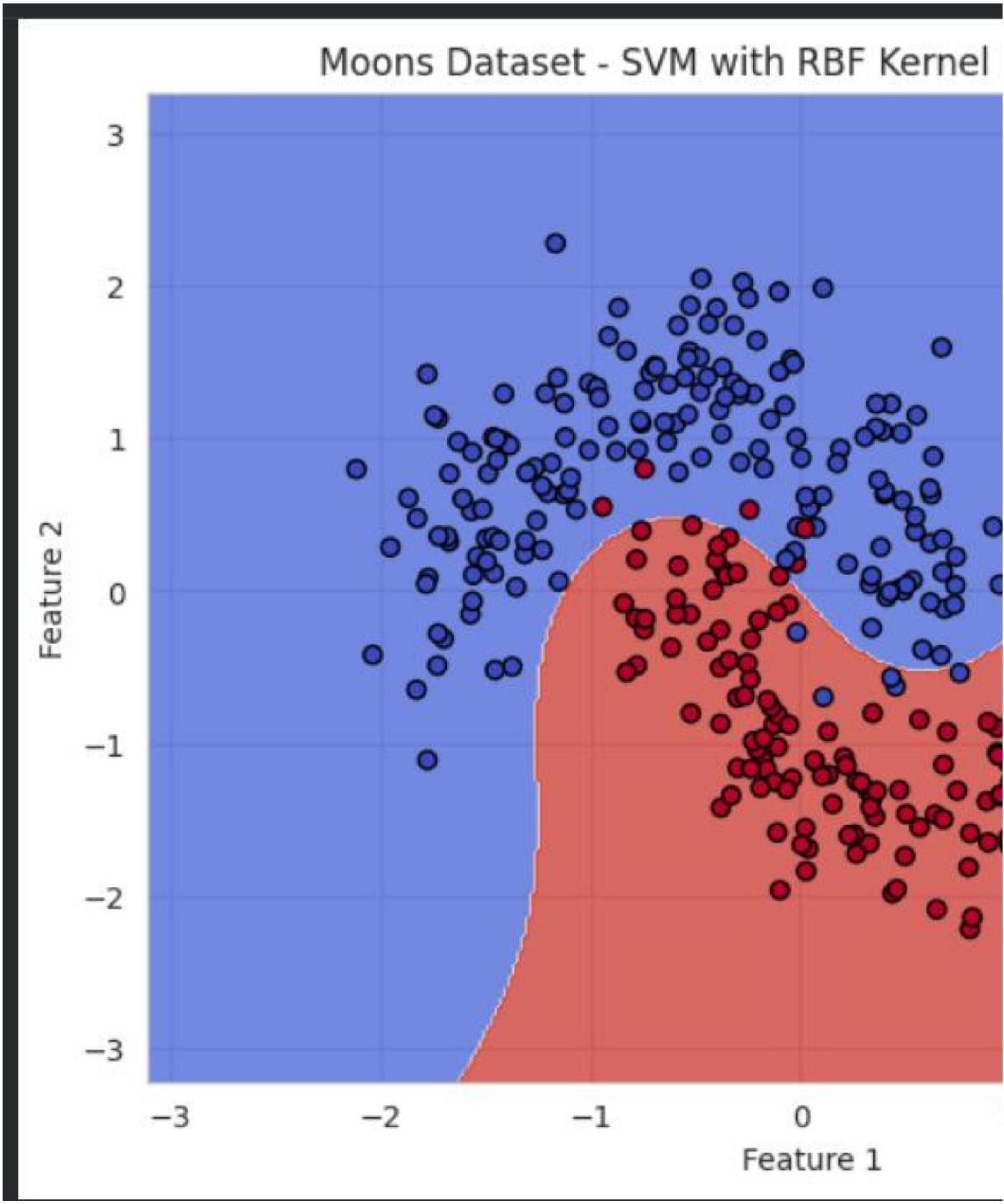
6. Results and Discussion

Moon Dataset

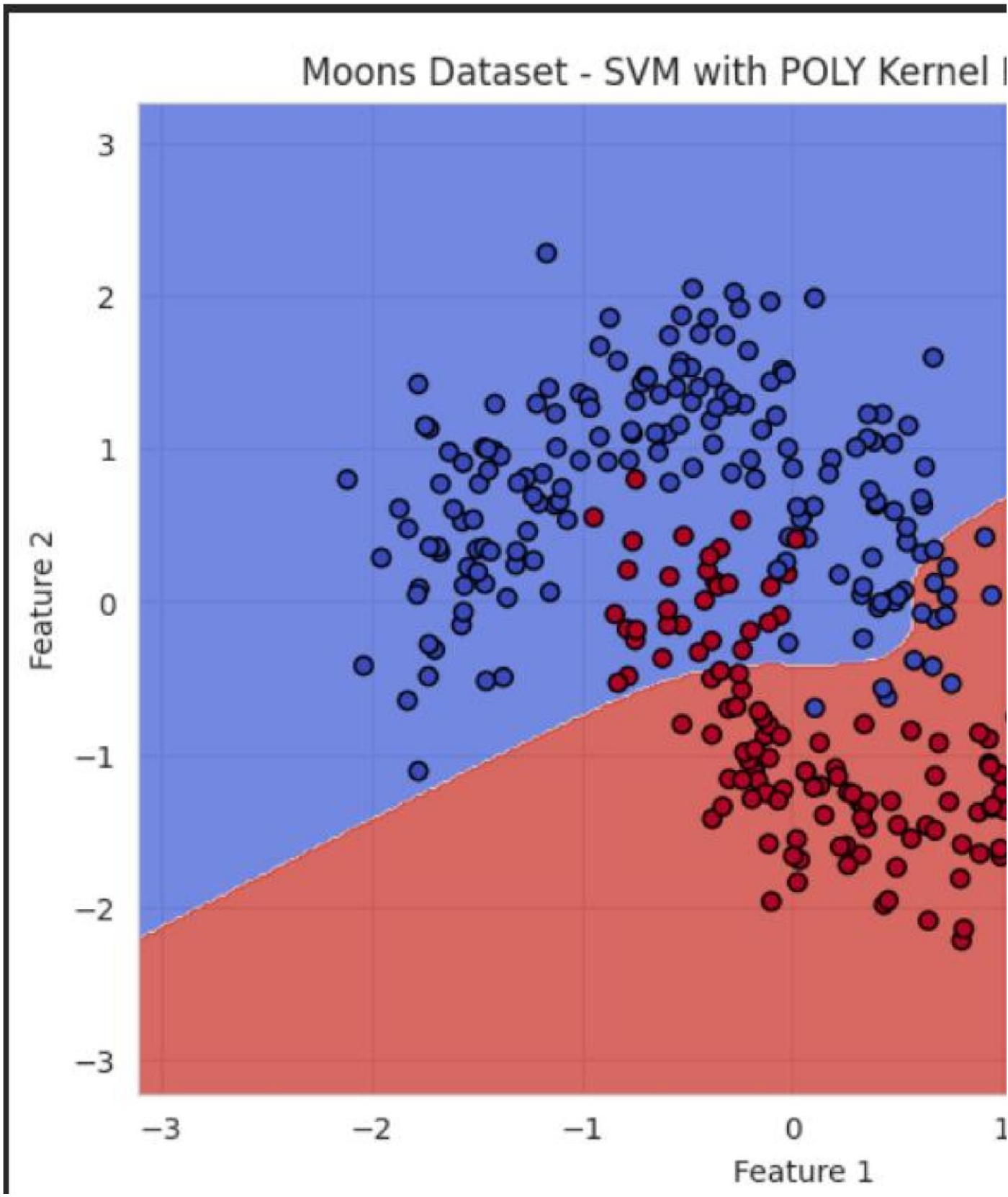


Kernal

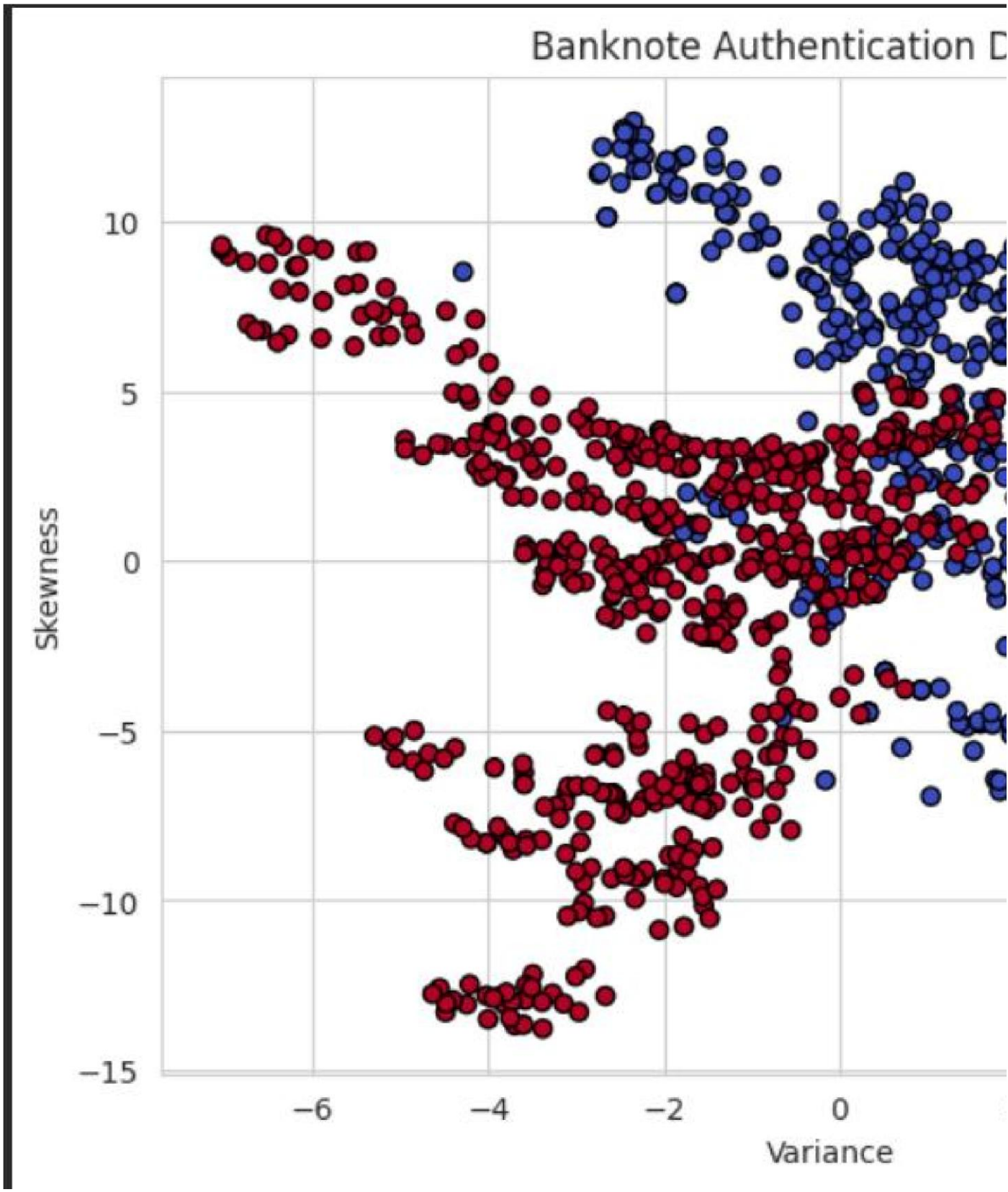

Moons Dataset - SVM with LINEAR Kernel
Feature 2
Feature 1
3
2
1
0
−1
−2
−3
−3
−2
−1
0
1


Moons Dataset - SVM with RBF Kernel
Feature 2
3
2
1
0
−1
−2
−3
−3
−2
−1
0
Feature 1

+


Moons Dataset - SVM with POLY Kernel
Feature 2
3
2
1
0
−1
−2
−3
−3
−2
−1
0
Feature 1

Banknote Authentication Dataset


Banknote Authentication D
10
5
0
−5
−10
−15
Skewness
−6
−4
−2
0
Variance

Train and Evaluate SVM Kernels

```
...  SVM with LINEAR Kernel YOUR_SRN
                  precision    recall  f1-sco

          Forged       0.90      0.88      0.
         Genuine       0.86      0.88      0.

        accuracy                           0.
       macro avg       0.88      0.88      0.
    weighted avg       0.88      0.88      0.

    ----------------------------------------

    SVM with RBF Kernel YOUR_SRN
                  precision    recall  f1-sco

          Forged       0.96      0.91      0.
         Genuine       0.90      0.96      0.

        accuracy                           0.
       macro avg       0.93      0.93      0.
    weighted avg       0.93      0.93      0.
```

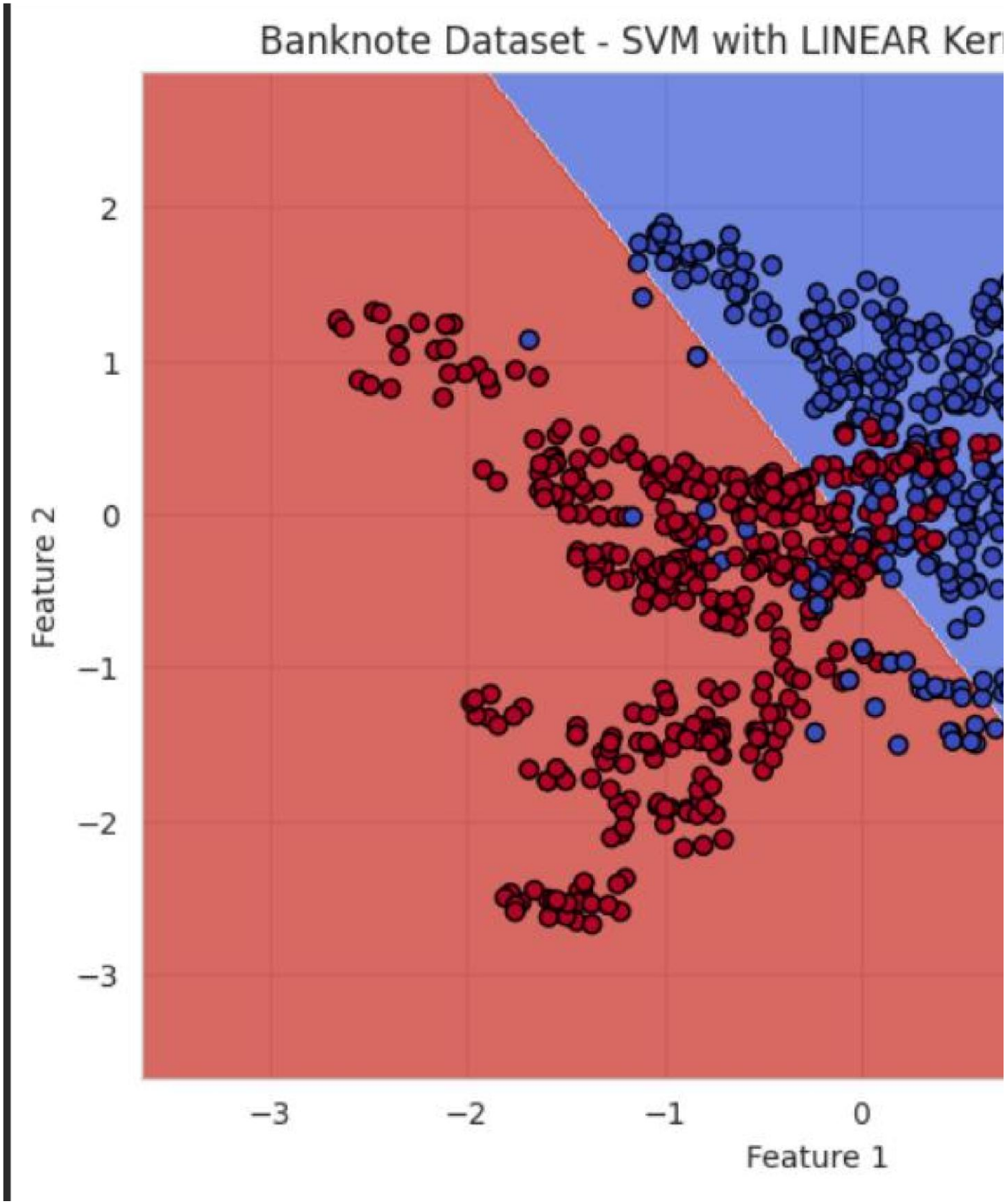
```
SVM with POLY Kernel YOUR_SRN
              precision    recall  f1-score   support

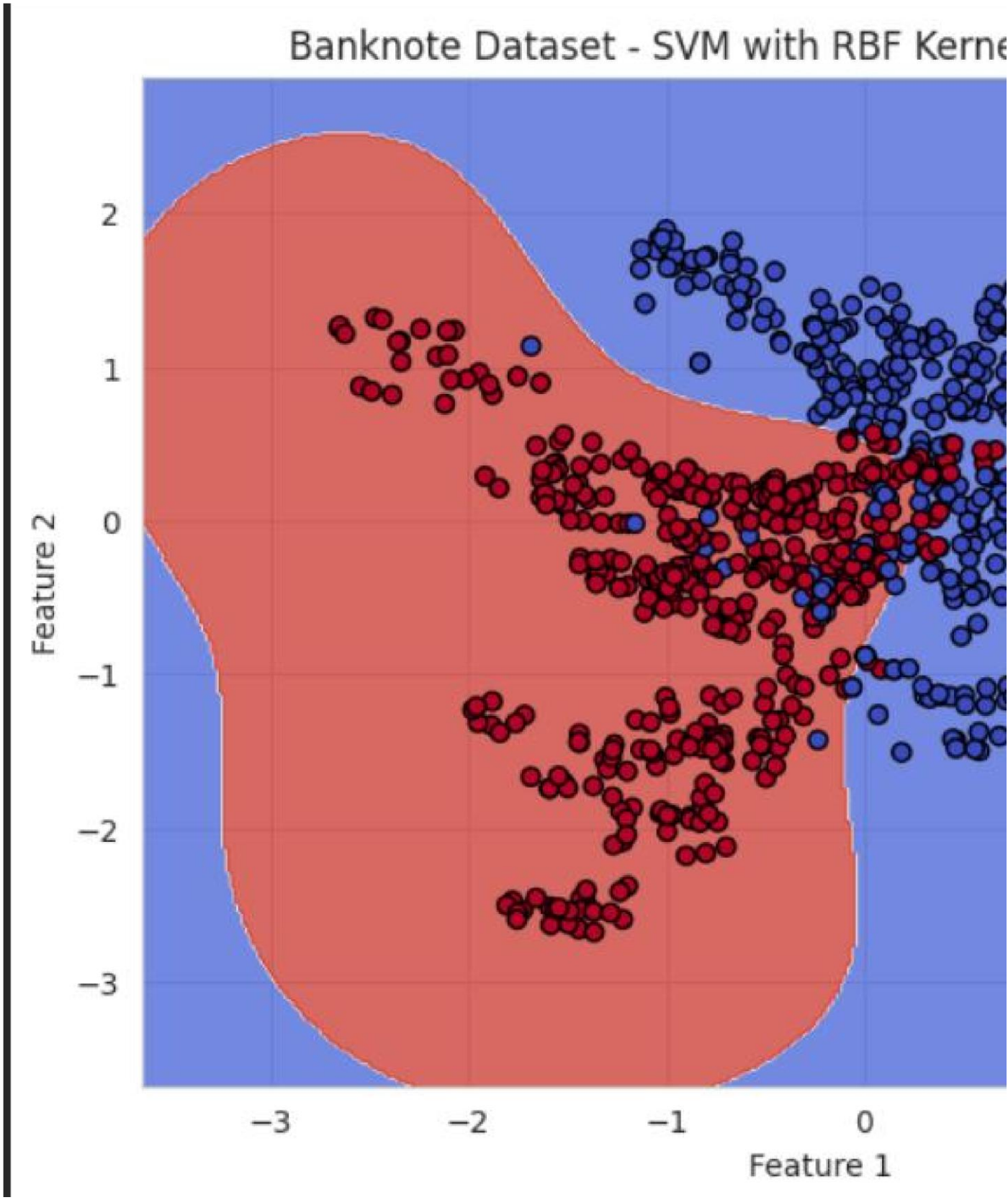
      Forged       0.82      0.91      0.87       229
     Genuine       0.87      0.75      0.81       183

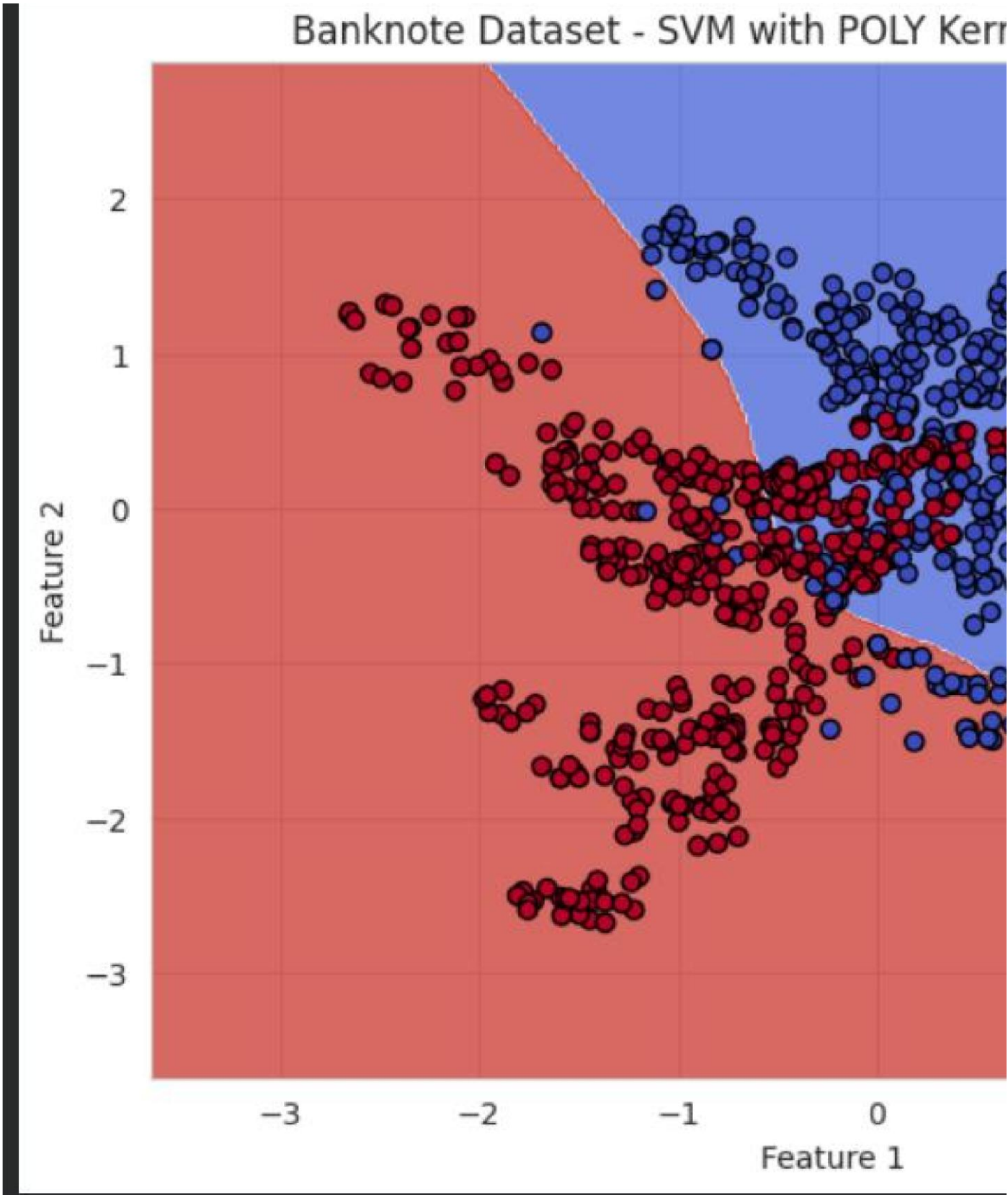
    accuracy                           0.84       412
   macro avg       0.85      0.83      0.84       412
weighted avg       0.85      0.84      0.84       412

----------------------------------------
```

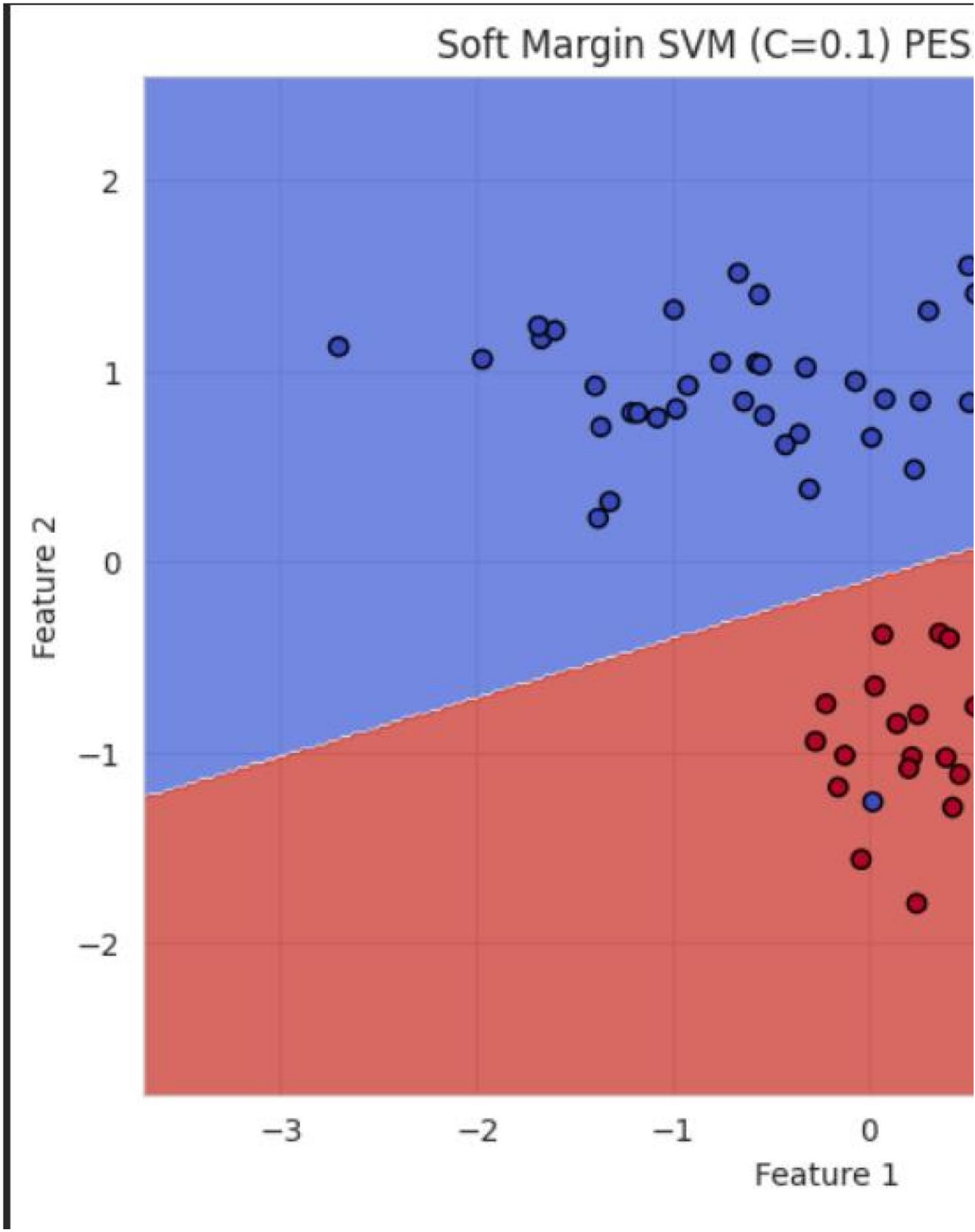
Visualize Decision Boundaries


Banknote Dataset - SVM with LINEAR Ker
Feature 2
2
1
0
−1
−2
−3
−3
−2
−1
0
Feature 1

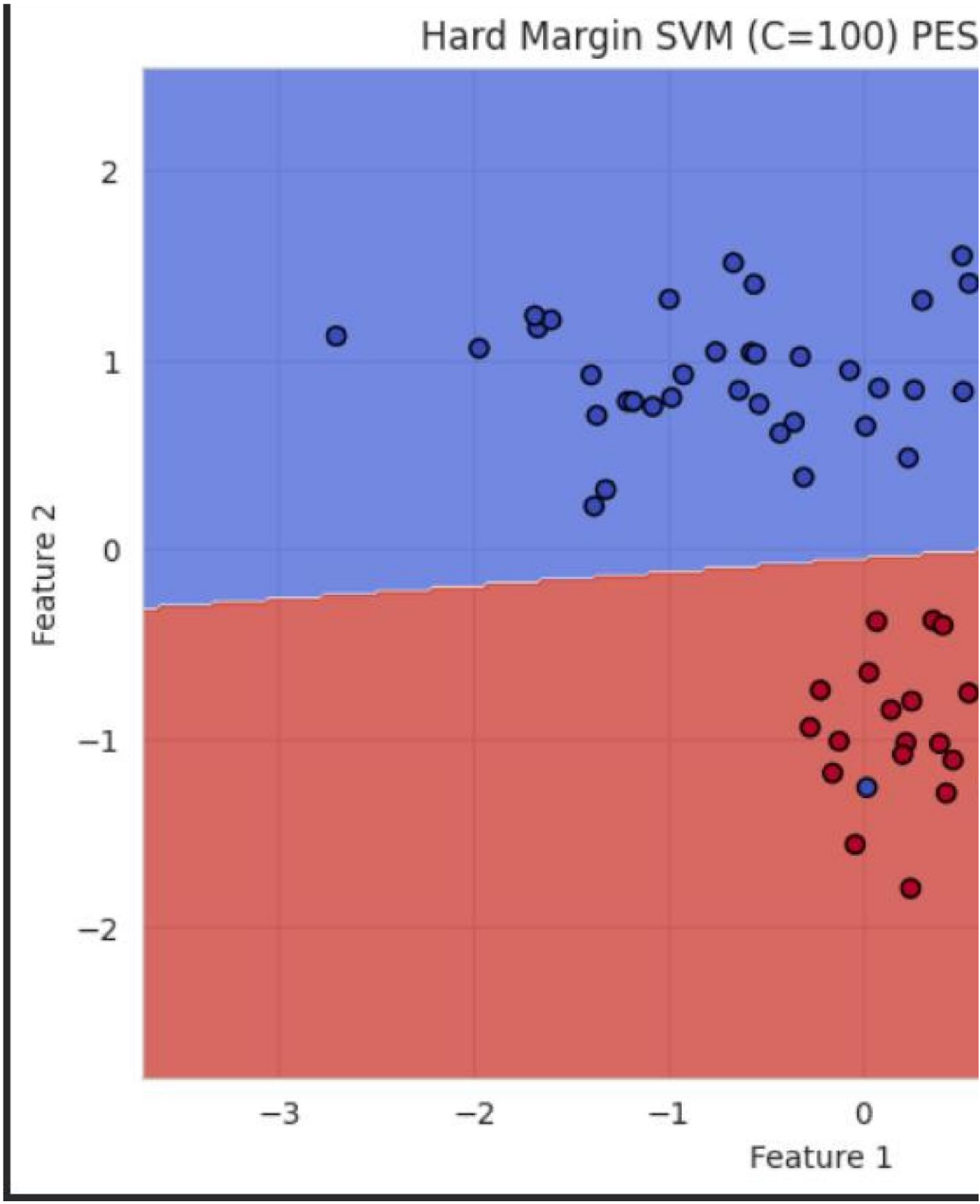

Banknote Dataset - SVM with RBF Kerne
Feature 2
2
1
0
−1
−2
−3
−3
−2
−1
0
Feature 1


Banknote Dataset - SVM with POLY Kern
Feature 2
2
1
0
−1
−2
−3
−3
−2
−1
0
Feature 1

Soft Margin:


Soft Margin SVM (C=0.1) PES
2
1
0
−1
−2
Feature 2
−3
−2
−1
0
Feature 1

Hard Margin


Hard Margin SVM (C=100) PES
Feature 2
2
1
0
−1
−2
−3
−2
−1
0
Feature 1

8. Conclusion

This lab demonstrates that SVM performance depends on the dataset, the kernel, and the value of C. Linear kernels create straight decision boundaries, while RBF and Polynomial kernels can model non-linear patterns. A lower C value allows more training errors in exchange for a wider margin, while a higher C value penalizes errors more strongly. The final model should be selected by comparing measured performance on unseen data. Before submission: replace the coverpage placeholders, add actual model results, and insert all 14 clearly labelled screenshots. Save the completed notebook with outputs visible